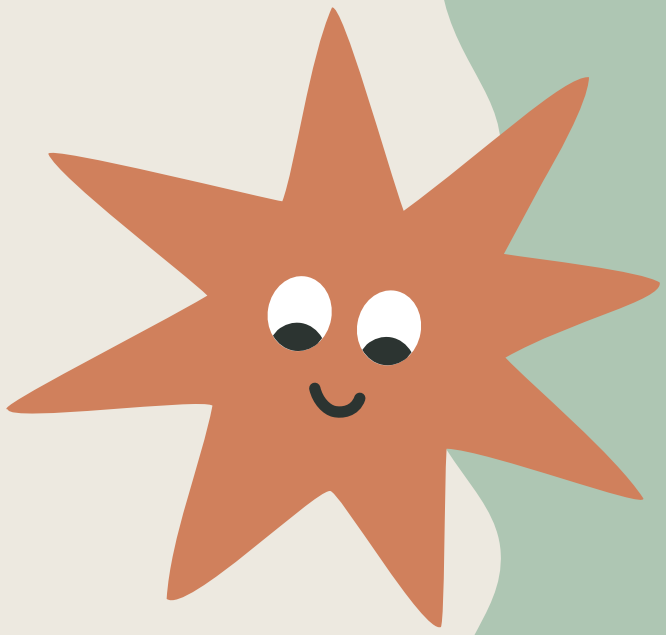
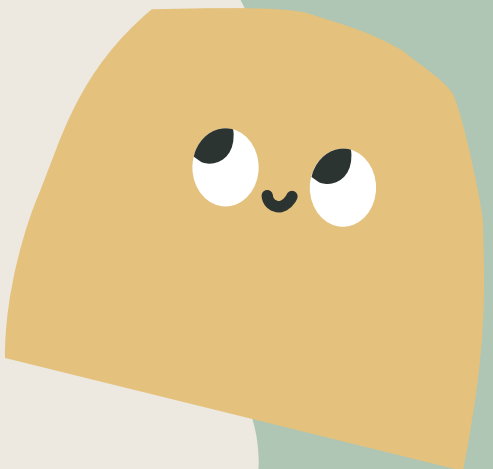
History and Evolution of Artificial Intelligence

Lecture 2





Learning Outcomes (Lecture 1)

➢At the end of this lecture, students will be able to:

- Explain the historical development of Artificial Intelligence.
- Describe the contribution of Alan Turing to AI.
- Explain the Turing Test and its purpose.
- Explain Symbolic AI and its underlying idea.
- Understand expert systems and rule-based reasoning.
- Explain why AI experienced periods known as **AI Winters**.
- Explain the Machine Learning Revolution.
- Explain why Deep Learning became successful.
- Understand the emergence of Generative AI.
- Identify important milestones in AI development.
- Recognize AI systems used in everyday life.

Prepared By: Ehtisham Rasheed

Why Study AI History?

- Today's AI is the result of **decades of research, failures, improvements, and technological breakthroughs**
- If we understand the history, we can understand:
  - why early AI used rules,
  - why those systems failed,
  - why machine learning became popular,
  - why large datasets became important,
  - why GPUs changed AI,
  - why deep learning became successful,
  - and why Generative AI emerged.
- So the history of AI is essentially the story of **how our approach to creating intelligent machines changed over time**.

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AI Before the Term "Artificial Intelligence"

- The idea of creating artificial beings is much older than computers
- Humans have imagined intelligent machines for centuries
- Examples include **mechanical automata**, **artificial beings in mythology**, and **programmable mechanical devices**
- However, **modern AI requires computers** because computers provide the ability to:
  - store information,
  - perform calculations,
  - execute algorithms,
  - process large amounts of data,
  - and automate decision-making.

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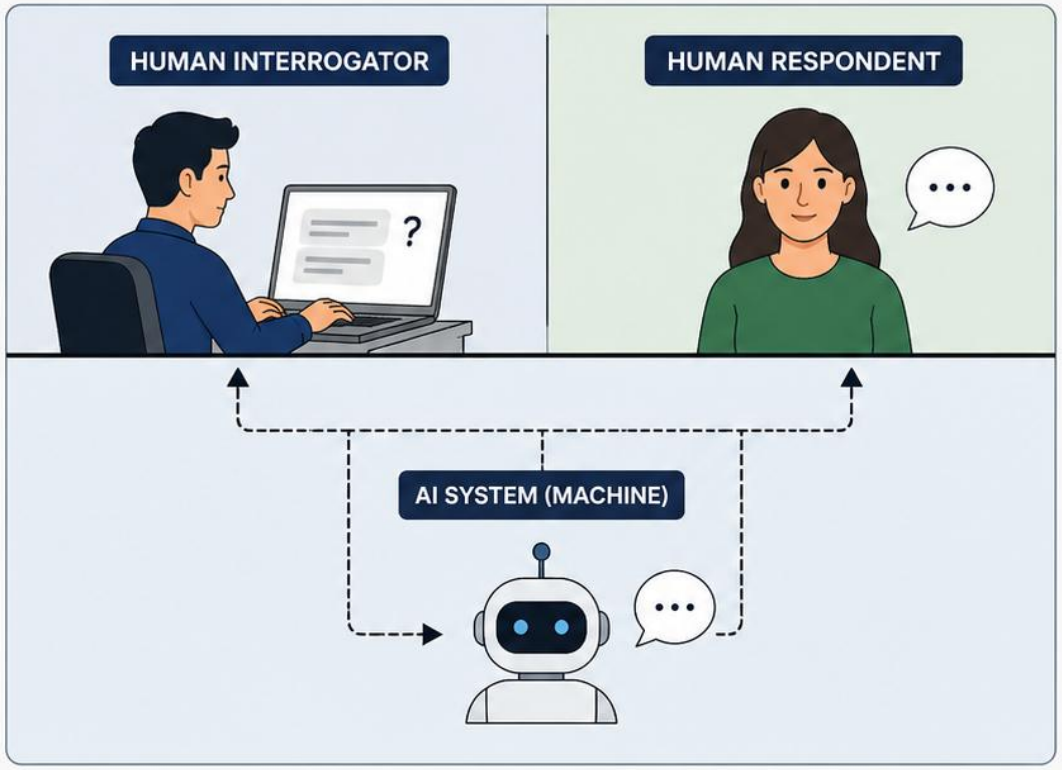
Alan Turing

- One of the most important names in AI history is: **Alan Turing**
- A British mathematician, logician and computer scientist
- He made fundamental contributions to:
  - Computation
  - Mathematics
  - Cryptography
  - Computer science
  - and the theoretical foundations of AI
- One of his most famous questions was essentially:
  - Can machines think?
  - Turing proposed a test called “Turing Test”

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TURING TEST

A test of a machine's ability to exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human.



HOW IT WORKS

1. A human interrogator communicates with two unseen entities (a human and a machine) using natural language (e.g., text).
2. The interrogator does not know which responder is the human and which is the machine.
3. The interrogator asks questions to both and receives answers.
4. If the interrogator cannot reliably distinguish the machine from the human, the machine is said to have **PASSED** the Turing Test.

OUTCOME



If the machine's responses are **indistinguishable** from the human's responses, it is considered to exhibit intelligent behavior.

KEY CONDITIONS

Natural Language Conversation
Communication must be in a natural language.

Hidden Identities
The interrogator cannot see or hear the responders.

Impersonation
The machine must try to imitate human-like intelligence.

KEY POINT



Proposed by Alan Turing in 1950, the test does not check internal thinking or consciousness; it only evaluates **observable behavior.**

> "I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted."
>
> – Alan Turing, 1950

Has AI Passed the Turing Test?

➢GPT-4 & GPT-4.5 (2024–2025):

- Achieved human-level deception in rigorous UC San Diego testing
- GPT-4 fooled judges 54% of the time
- GPT-4.5 fooled judges 73% of the time, outperforming real humans

➢Eugene Goostman (2014)

- First bot to cross Alan Turing's historic 30% deception threshold
- Simulated a 13-year-old Ukrainian boy to excuse poor grammar
- Critics dismissed the win as a clever loophole rather than true intelligence

➢ELIZA (1966)

- A primitive script mimicking a psychotherapist via pattern matching

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The Birth of Artificial Intelligence as a Field

- In 1956, the Dartmouth Summer Research Project on Artificial Intelligence helped establish AI as a formal research field
- John McCarthy proposed the term “Artificial Intelligence”
- The researchers believed that aspects of human intelligence might be described precisely enough that machines could simulate them
- They believed that computers might soon be able to:
  - solve difficult problems,
  - reason logically,
  - understand language,
  - play games
  - and eventually demonstrate general intelligence

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The Birth of Artificial Intelligence as a Field

- This period produced several important ideas and systems
- This period launched major research into search, reasoning, problem solving, language, and game playing
- Computers at that time were **extremely limited compared with modern computers**
- They had:
  - limited memory
  - limited processing power
  - limited storage
  - and very little data

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Symbolic AI

- It is one of the dominant approaches during early AI research
- Also called
  - Symbolic Artificial Intelligence
  - Logic-based or Rule-based
  - GOFAI (Good Old-Fashioned AI)
- A symbol represents real world object, concept, fact or relationship
- Symbolic AI represent knowledge using:
  - Symbols, rules, logic, facts or relationships

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How Symbolic AI Works

➢Consider this simple example

```
Fact:
Ali is a student.

Fact:
All students are learners.

Rule:
If someone is a student,
then that person is a learner.
```

The AI system can reason

```
Ali is a student
        +
Students are learners
        ↓
Ali is a learner
```

➢This is called **rule-based reasoning**.

➢Important idea is:

- Humans explicitly provide knowledge and rules to the computer

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HOW SYMBOLIC AI WORKS

Symbolic AI represents knowledge as symbols (facts) and rules.
The system uses logic and reasoning to derive new conclusions.

1. KNOWLEDGE (FACTS)

Information about the world is stored as facts (symbols).

- Fact 1: Ali has fever.
- Fact 2: Ali has cough.
- Fact 3: Ali has runny nose.
- Fact 4: Ali's temperature is 39°C.

Facts are stored in the **KNOWLEDGE BASE.**

2. RULE BASE

Domain knowledge is encoded as IF–THEN rules.

Rule 1:
IF temperature > 38°C
THEN fever.

Rule 2:
IF cough AND runny nose
THEN cold.

Rule 3:
IF fever AND cough
THEN possible infection.

Rule 4:
IF possible infection AND temperature > 38°C
THEN recommend doctor.

3. INFERENCE ENGINE (REASONING)

Applies rules to facts to infer new information.

Reasoning Process

Step 1: Check Rule 1
Temperature is 39°C → Fever ✓

Step 2: Check Rule 2
Cough AND Runny nose → Cold ✓

Step 3: Check Rule 3
Fever AND Cough → Possible Infection ✓

Step 4: Check Rule 4
Possible Infection AND Temperature > 38°C → Recommend Doctor ✓

4. CONCLUSIONS (NEW FACTS)

New information (conclusions) is derived from facts and rules.

- **Conclusion 1:** Ali has fever.
- **Conclusion 2:** Ali has cold.
- **Conclusion 3:** Ali has possible infection.
- **Conclusion 4:** Recommend Doctor.

Conclusions can be stored as new facts and used for further reasoning.

OVERALL PROCESS

Facts (Symbols) from the world → Rules encode human expert knowledge → Inference Engine applies logic and reasoning → New conclusions are generated → New facts can be used for more reasoning

EXAMPLE (MEDICAL DIAGNOSIS)

Input Facts
Fever, Cough, Runny Nose, Temp = 39°C

→ **Rules**
IF–THEN rules from medical expert

→ **Reasoning**
Apply rules step by step

→ **Output (Conclusions)**
Fever, Cold, Possible Infection, Recommend Doctor

KEY POINT

Symbolic AI does not learn from data. It reasons using explicit symbols, facts and rules provided by humans.

Early AI Programs

- Logic Theorist (1950s)
  - It attempted to prove mathematical theorems
- General Problem Solver
  - It explored general strategies for solving problems
- ELIZA (1960s)
  - ELIZA was an early natural-language conversation program
  - It used pattern matching to simulate conversation
- These systems demonstrated that computers could perform tasks that appeared intelligent in restricted settings

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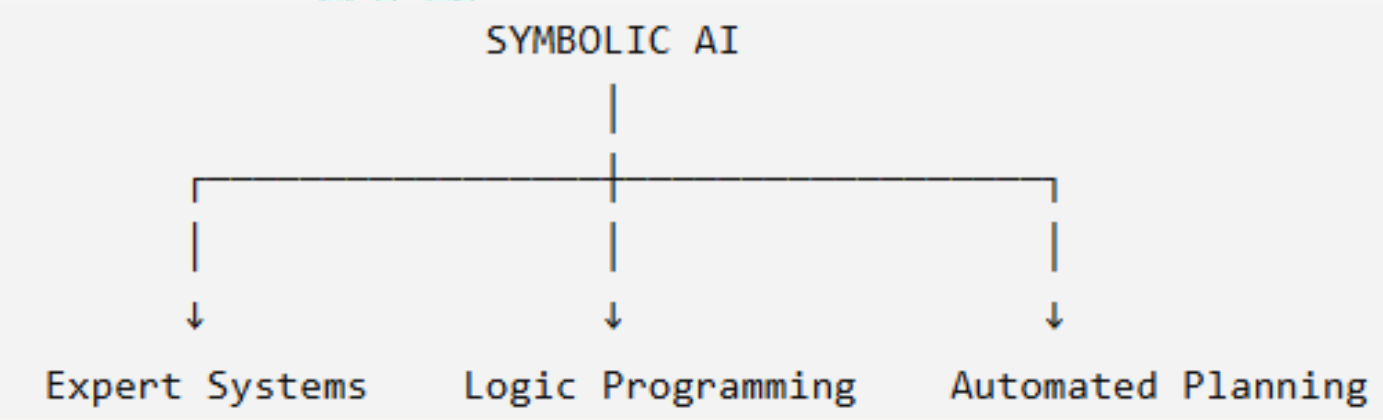
What is an Expert System?

- Another major development in AI was: Expert Systems
- An expert system solves problems in a specialized domain using knowledge and rules from human experts
- Knowledge Base → stores facts and rules
- Inference Engine → applies rules to facts and derives conclusions
- Famous Expert Systems:
  - **MYCIN (1970):** It was designed to assist with diagnosis and treatment recommendations for certain bacterial infections
  - **XCON (1980):** An expert system used to configure computer systems

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Symbolic AI vs Expert System

- **Symbolic AI** is an approach to AI where knowledge is explicitly represented using:
  - Symbols, facts, rules, logic, relationship
  - And the system uses reasoning/inference to reach conclusions
- An **Expert System** is an AI application designed to solve problems in a **specific domain** using knowledge obtained from human experts
  - Expert systems are one important application of symbolic AI



Prof. Dr. Ehtisham Rasheed

Problems with Symbolic AI and Expert Systems

1. Manually writing rules for every possible situation becomes extremely difficult

- Imagine that we want to create a system capable of recognizing cats
- We might try to write rules:

```
IF animal has two eyes
AND four legs
AND pointed ears
AND tail
THEN cat
```

- Dog also has these features
- So, we've to add more rules. But then we encounter:
  - Different cat breeds, colors, positions, partial visibility, lighting changes
  - Soon we have thousands or millions of possible rules
  - It is extremely difficult to manually encode all the knowledge required for real-world situations

Prepared By: E Gashed

Problem 1: Manually writing rules for every possible situation becomes extremely difficult

In Symbolic AI / Expert Systems, humans must write rules for every possible scenario.
As the number of possibilities grows, the number of rules explodes!

EXAMPLE

A simple medical diagnosis system considers only 4 symptoms:

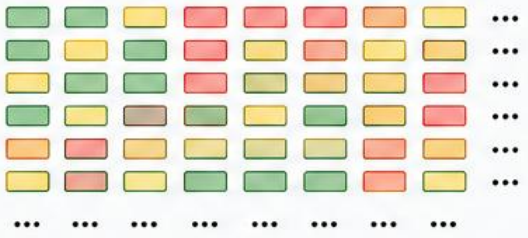


Fever Cough Headache Fatigue

If each symptom can be:

Present | Absent | Unknown

Possible combinations = $3^4 = 81$



We need a rule for each possible combination to handle every case!

RULE EXPLOSION

Businesses rules needed

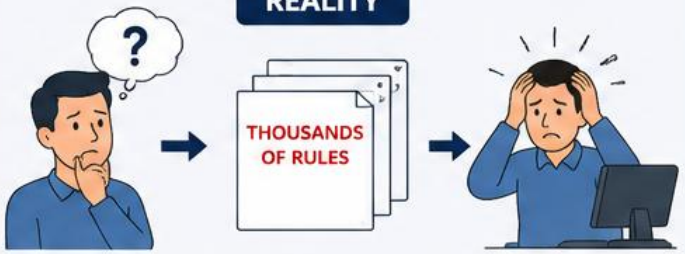
No. of Symptoms	Possible Combinations ($3^{\wedge n}$)
4	$3^4 = 81$
6	$3^6 = 729$
8	$3^8 = 6{,}561$
10	$3^{10} = 59{,}049$
12	$3^{12} = 531{,}441$
15	$3^{15} = 14{,}348{,}907$
20	$3^{20} = 3{,}486{,}784{,}401$

Even for a small number of factors, the number of rules becomes unmanageable!

WHY IT BECOMES EXTREMELY DIFFICULT

- Experts cannot think of every possible combination.
- Writing thousands or millions of rules is time-consuming and error-prone.
- Rules become hard to organize, maintain and update.
- Adding a new symptom or condition multiplies the number of rules.
- System may still miss rare or unseen situations.

REALITY



Difficult to build, maintain and keep complete!

KEY TAKEAWAY

Manually writing rules does not scale.
This limitation led to the need for **Machine Learning**, where systems learn patterns from data instead of relying on humans to write every single rule.

2. The Knowledge Acquisition Problem

- This problem became known as the:
  - Knowledge Acquisition Bottleneck
- A **bottleneck** is a point that limits the overall performance or progress of a system
- An expert may know how to solve a problem but may not be able to clearly explain every rule they use
- Also called **“Tacit knowledge”**

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Problem 2: The Knowledge Acquisition Problem

Expert systems depend on knowledge from human experts.
But getting that knowledge is extremely difficult, slow, expensive and often incomplete.

HOW KNOWLEDGE IS ACQUIRED

Human Expert



Has knowledge and experience in the domain

↓

Knowledge Engineer (Interviewer)



Extracts knowledge by asking questions, observing and discussing

↓

Knowledge Base (Facts + Rules)



Knowledge is written as facts and rules for the system

↓

Expert System



Uses this knowledge to make decisions and give advice

WHY IT IS A MAJOR PROBLEM



TIME CONSUMING
Experts are busy and hard to reach. Extracting knowledge takes a lot of time.



EXPENSIVE
Involves skilled knowledge engineers and many sessions with experts.



DIFFICULT TO EXPRESS
Experts have knowledge in their heads, but cannot always explain it clearly.



TACIT KNOWLEDGE
Much of the knowledge is tacit (experience-based) and not easily verbalized.



INCOMPLETE KNOWLEDGE
Experts may forget to mention important things. Some knowledge may remain missing.

EXAMPLE: MEDICAL DIAGNOSIS SYSTEM



Doctor (Expert)



Knowledge Engineer asks questions

RULES
1. IF fever AND cough THEN infection
2. IF ...
3. IF ...
...

Rules are written in the system

Challenges in this process:

- Doctor may not have time.
- Some knowledge is inexperience and not easily explained.
- Different doctors may have different opinions.
- Important rules may be missed.

CONSEQUENCES



The system may give wrong or incomplete advice.



System needs frequent updates and corrections.



Hard to maintain as knowledge grows.



Limits the reliability and effectiveness of expert systems.

KEY TAKEAWAY

The success of an expert system depends on the quality of knowledge it has,
but acquiring that knowledge from human experts is one of the biggest challenges.



3. The Real World is Complicated

- Real-world environments are often:
  - Uncertain
  - Incomplete
  - Noisy
  - Changing
  - Unpredictable
- Symbolic systems often performed well in controlled environments

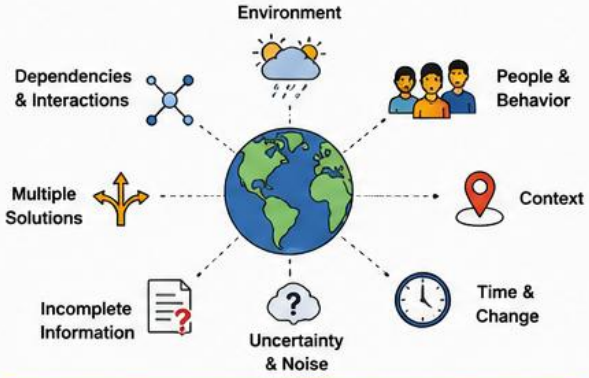
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Problem 3: The Real World is Complicated

Symbolic AI and Expert Systems assume the world is simple and well-structured.
But the real world is uncertain, noisy, dynamic and full of exceptions.

1. COMPLEXITY OF REAL WORLD

A real-world situation has many interacting factors, hidden conditions and dependencies.



⚠ The number of possible situations is enormous and cannot be fully captured by rules.

2. UNCERTAINTY AND IMPRECISE INFORMATION

Real-world data is often uncertain, vague or incomplete.



"I have slight fever" not very high." — **Rules need exact conditions, but real world is not exact.**



"The pain is kind of sometimes." — **Vague, subjective and linguistic information.**



"Maybe it is due to stress or tiredness." — **Uncertainty makes it hard to apply strict rules.**

3. DYNAMIC AND CHANGING ENVIRONMENT

The real world changes constantly.



- New situations appear that were never seen before.
- People's behavior, rules, laws and environments change.
- A fixed set of rules quickly becomes outdated.

4. EXCEPTIONS ARE EVERYWHERE

In real life, there are always exceptions to rules.

Rule:
IF a bird has wings
THEN it can fly.

Exceptions:
- Penguin
- Ostrich
- Injured birds



✗ It is almost impossible to list all exceptions in rules.

5. LARGE NUMBER OF INTERACTIONS

When many factors interact, behavior becomes non-linear and hard to predict.



✗ Rule-based systems struggle with such complexity.

REAL WORLD EXAMPLE

Consider a self-driving car.



- ✓ Different weather
- ✓ Different roads
- ✓ Different drivers
- ✓ Sudden events
- ✓ Unknown objects
- ✓ Endless situations

Writing rules for all these situations is impossible!

KEY TAKEAWAY

The real world is uncertain, dynamic, noisy and full of exceptions.
Symbolic AI and Expert Systems, based on fixed rules, find it very hard to handle such complexity.
This limitation motivated the shift towards Machine Learning, where systems learn from data instead of trying to manually cover every possible situation.



AI Winters

- It is a period of reduced funding, interest, and confidence in AI research
- First major AI Winter: roughly 1974–1980
- Second AI Winter: roughly late 1980s to early/mid-1990s
- Common causes: ambitious promises, limited computing power, difficult problems, and disappointing results
- AI research continued during these periods, eventually leading to new approaches

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The Shift Toward Machine Learning

- The fundamental idea changed
- Symbolic AI (Old Approach)
  - Tell the computer the rules
  - Human → Rules → Computer → Decision
- Machine Learning (New Approach)
  - Give the computer data and allow it to learn patterns
  - Data → Learning Algorithm → Model → Prediction
- Example: spam detection can learn from thousands of labeled emails
- The learned model can then be applied to new, unseen data

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The Three Major Types of Machine Learning

- Supervised Learning
  - The system learns from examples where the correct answer is known
  - Example: Email → Spam / Not Spam
- Unsupervised Learning
  - The system receives data without predefined answers and tries to discover patterns or groups
  - Example: Customer Data → Clustering → Customer Groups
- Reinforcement Learning
  - The system learns through interaction with an environment using **rewards and penalties**
  - A game-playing AI is a common example

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Machine Learning Revolution

➢1990s onward, machine learning became increasingly important due to:

1. More Data
   - The internet generated enormous amounts of digital information
2. Faster Computers
   - Computational power increased significantly
3. Better Algorithms
   - Researchers developed improved learning methods
4. Better Storage
   - Large datasets became easier to store
5. Improved Software
   - Libraries and frameworks made machine learning easier to implement

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IBM Deep Blue

- In 1997 **Deep Blue** defeated world chess champion **Garry Kasparov**
- It demonstrated that a computer could defeat one of the world's strongest chess players
- However, Deep Blue was not a modern deep-learning system
- Deep Blue relied heavily on search, chess-specific knowledge, evaluation functions, and specialized hardware
- Historical lesson:
  - AI can be powerful using classical search and knowledge

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AI Enters the Data Era

- As the internet expanded, enormous datasets became available
- Examples include:
  - Photographs, videos, documents, web pages, product reviews, social media posts
- Instead of manually telling computers everything, we could let them learn from huge amounts of data
- But another problem remained
  - Traditional machine-learning methods had limitations when dealing with extremely complex data
- This led to the next major revolution

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Deep Learning Revolution

- Deep Learning is a subfield of machine learning based primarily on **multi-layer neural networks**
- "Deep" refers to having multiple computational layers
- Deep neural networks can learn complex representations directly from large datasets
- Major progress accelerated when big data, powerful GPUs, improved algorithms, and better software came together

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AlexNet (2012) and Image Recognition

- AlexNet achieved a major breakthrough in large-scale image classification in 2012
- This event is often considered a major turning point in modern AI
- Basic idea is:
  - Millions of images → Deep Neural Network → Learn Visual Patters → Image Classification
- Instead of manually programming every visual feature, the neural network learned useful representations from data
- Modern computer vision accelerated rapidly after this milestone

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AI Becomes Mainstream

- During the 2010s, AI started becoming part of everyday products
- Examples include
  - Voice assistants
  - Face recognition
  - Recommendation systems
  - Automatic translation
  - Fraud detection
  - Autonomous driving research
- AI moved from research laboratories into consumer products and business applications

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AlphaGo (2016)

- Developed by Google DeepMind
- AlphaGo defeated **Lee Sedol**, one of the world's strongest Go players
- Go has an enormous number of possible game positions
- The number of possibilities is far larger than in chess
- AlphaGo combined neural network, search, and reinforcement learning
- This demonstrated the power of combining multiple AI techniques

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The Transformer (2017)

- Shift from deep learning to modern generative AI happened with a paper published by Google researchers: *"Attention Is All You Need."*
- **The Invention:** They introduced the **Transformer architecture**
- Instead of reading text word-by-word (like older deep learning models), Transformers use **Self-Attention**
- They look at an entire document at once, understanding the context and relationships between words regardless of distance
- Transformers allowed for massive parallel processing, meaning engineers could finally train AI on the entire public internet

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The Rise of Large Language Models (2018–2022)

- A **language model** is an AI model trained to understand patterns in language and predict or generate language-related outputs
- **Large Language Models (LLMs)** are advanced artificial intelligence programs trained on massive amounts of text to understand, summarize, generate, and predict new content
- Common examples of LLMs
  - **GPT-4o / GPT-4:** Developed by OpenAI
  - **Claude 3 / 3.5:** Developed by Anthropic
  - **Gemini:** Developed by Google
  - **Llama 3:** Developed by Meta (open-source)

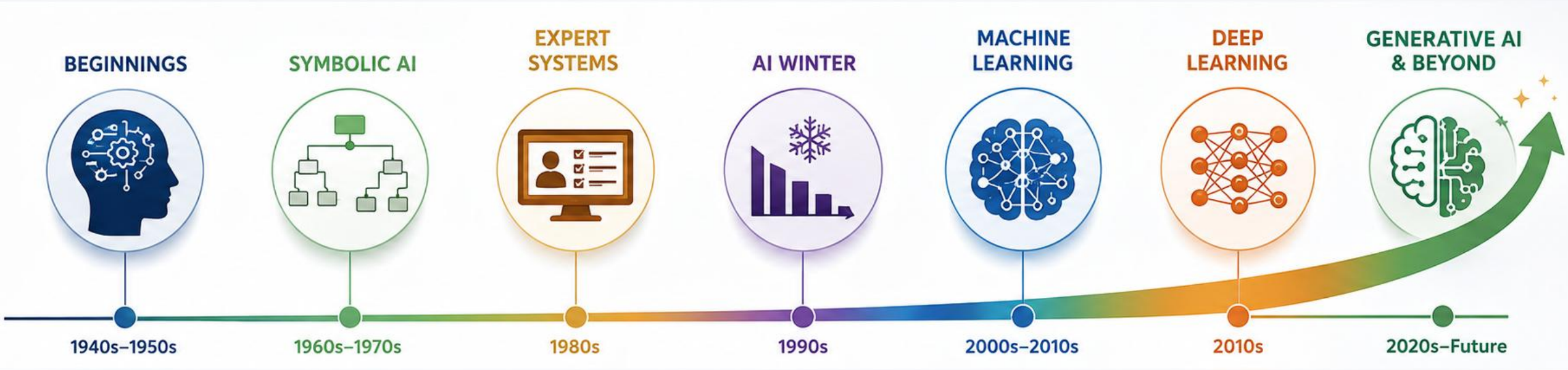
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The Generative AI Era (2022 – Present)

- Generative AI refers to AI systems capable of generating new content
- Examples:
  - Text
  - Images
  - Audio
  - Video
- The rapid growth of generative AI has made AI accessible to millions of ordinary users
- The public release of **ChatGPT in November 2022** became a major milestone in the widespread adoption of generative AI

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THE EVOLUTION OF AI



Exploring the Possibilities → Building the Foundations → Applying Knowledge to Solve Problems → Facing Challenges & Limitations → Learning from Data → Power of Deep Neural Networks → New Era of Intelligence